

A High-Resolution Diffraction and Spectroscopic Study of the Low-Temperature Phase Transformation of Hexagonal to Tetragonal GeO₂ with and without Alkali Hydroxide Promotion

Thomas Bielz,[†] Soipatta Soisuwan,^{†,§} Reinhard Kaindl,[‡] Richard Tessadri,[‡] Daniel M. Többsen,[‡] Bernhard Klötzer,[†] and Simon Penner^{*,†}

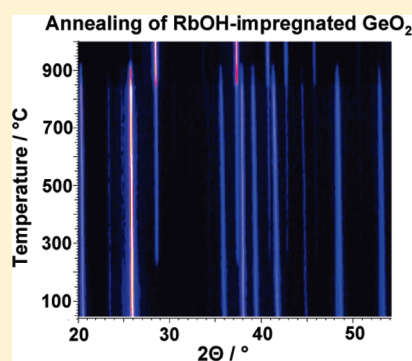
[†]Institute of Physical Chemistry, University of Innsbruck, Innrain 52a, A-6020 Innsbruck, Austria

[‡]Institute of Mineralogy and Petrography, University of Innsbruck, Innrain 52d, A-6020 Innsbruck, Austria

[§]Department of Chemical Engineering, Burapha University, T-20131 Chonburi, Thailand

S Supporting Information

ABSTRACT: Pure and RbOH-impregnated hexagonal GeO₂ were studied by high-resolution and -temperature X-ray diffraction, thermal gravimetric analysis coupled with mass spectrometry, and high-temperature Raman spectroscopy in the temperature range 313–1300 K for a direct comparison of the phase transformation from hexagonal to tetragonal GeO₂. On both samples, the phase transformation was observed to follow through distinct reaction steps. At low temperatures, a partial transformation to tetragonal GeO₂ occurred on both materials likely induced by the presence of strongly bound water. The removal of the latter is accompanied by substantial anisotropic lattice distortions and yields at maximum ~12% tetragonal GeO₂. This partial phase transformation appears to be almost independent of the alkali hydroxide addition. In contrast, the final high-temperature phase transformation around 1100 K results in full conversion to tetragonal GeO₂ on the RbOH-impregnated GeO₂ material, whereas, on pure hexagonal GeO₂, only partial transformation to the tetragonal phase was obtained.



1. INTRODUCTION

In the last decades, germanium dioxide (GeO₂) has evolved as a promising candidate for applications especially in optoelectronic devices.^{1–4} Although properties like a high dielectric constant and refractive index or a high thermal stability mainly determine the abilities for these applications,⁴ one of the most intriguing properties of GeO₂ is related to its polymorphism. It is well-known that GeO₂ exists in three main crystallographic phases at ambient temperatures, that is, in a rutile-type (tetragonal, space group *P4₂/mnm*) structure, an α -quartz (trigonal or hexagonal, space group *P3₁21*) structure, and an amorphous glassy state.^{5–8} Most important regarding the structure of crystalline phases, the tetragonal modification exhibits octahedrally, the hexagonal structure tetrahedrally coordinated Ge.¹ Both of the phases accordingly differ in their chemical properties, especially concerning solubility in water (the tetragonal phase is insoluble, the hexagonal phase slightly soluble in water).¹ The transformation temperature of the rutile into the α -quartz phase is 1306 K.⁹

In order to study the physicochemical properties of the different phases, several methods have been proposed and applied to catalyze the transformation from the hexagonal to the tetragonal modification.^{10–15} Without catalyst, it was stated that pure germanium dioxide does not undergo a phase transformation upon heating up to 1300 K.¹⁶ However, immediate formation of a low amount of tetragonal GeO₂ upon heating the

hexagonal modification at 653 K and above at 10–30 kbar pressure has been reported, alongside inhibition of further growth of the rutile-type phase by encapsulating the grains of the hexagonal phase, subsequently causing internal strain.¹⁷ Full transformation into the rutile-type modification has also been achieved by prolonged heating of the hexagonal phase at 1023 K for 700 h¹³ or by applying high pressures and temperatures (e.g., 2 kbar at 1000 K for 48 h).^{14,18} More chemically oriented methods include the use of crystallization aids such as ammonium fluoride¹⁹ or alkaline compounds, which catalyze the structural rearrangement from the 4-fold tetrahedral to the 6-fold octahedral Ge coordination at high temperatures.^{1,11–13,16} Especially the latter provide a convenient and rather easy access to the tetragonal GeO₂ modification. By far, the most widely used dopants in this respect are the carbonates of Li, Na, and K,^{1,11,12,16} less so Rb¹³ and Cs.¹¹ There is common agreement that the phase transformation proceeds via the intermediate formation of a crystalline or glassy germanate compound and that the kinetics of this phase transformation is significantly affected by the experimental conditions (e.g., gas atmosphere, water vapor, catalyst loading).^{11–13,16,20} Activation energies between

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170 and 400 kJ/mol depending on the catalyst used and the experimental parameters have been reported.²¹

Consequently, the research concerning this phase transformation has been mainly focused on high temperatures (above ~ 1073 K) and, therefore, basically on the mechanism of germanate glass formation. A large amount of data covering many aspects of this phase transformation has in turn been compiled over the last 40 years. What is missing, however, is a detailed investigation of the phase transformation behavior of pure hexagonal and alkali-doped hexagonal GeO_2 by high-resolution and temperature diffraction and spectroscopic techniques especially at low temperatures between 300 and 673 K. This will help to eventually clarify the discrepancies outlined above. We are thus able to eventually detect structural and/or chemical similarities in the phase transformation both between pure and RbOH-doped hexagonal GeO_2 but also at different low and high temperatures. Note that the term “pure” in the context of this contribution refers to a structurally homogeneous hexagonal GeO_2 sample without RbOH impregnation (i.e., without tetragonal phase). As will be clear from the discussion below, also this “pure” GeO_2 sample contains a number of OH-related impurities, which will give rise to the features observed in this work.

This task of a detailed study of the transformation between the hexagonal and tetragonal GeO_2 modifications with and without RbOH addition in the temperature range 300–1300 K is tackled by using high-resolution and -temperature X-ray diffraction (XRD), high-temperature Raman spectroscopy, and thermal gravimetric analysis (DTA-TG) coupled with mass spectrometry. X-ray photoelectron spectroscopy (XPS) was used for chemical analysis.

2. EXPERIMENTAL SECTION

2.1. Materials. Hexagonal GeO_2 powder (Alfa Aesar, 99.9999%) was used as such. For preparation of the doped hexagonal GeO_2 modification, hexagonal GeO_2 was impregnated with RbOH (RbOH aqueous solution 50%; 99.6% metals basis, Alfa Aesar) in a molar ratio of 99.995:0.005 and dried at 373 K for 12 h. Subsequently, both the doped and undoped samples were subjected to different heating routines (i.e., heating rates and times) in the temperature region 300–1300 K to monitor the kinetics of the phase transformation between the two modifications.

2.2. X-ray Diffraction. Ex situ powder X-ray diffraction experiments were performed under ambient conditions using a Siemens D5000 diffractometer and Cu $K\alpha$ radiation (1.54178 Å) at 300 K in the 2θ range 10° – 70° . High-temperature PXRD data for both phases were obtained using a Siemens D5005 diffractometer also using Cu $K\alpha$ radiation with parallel beam optics, scintillation counter, and theta–theta coupling geometry. Diffractograms were taken in the same 2θ range 10° – 70° in 0.02° steps and with a 2 s/step counting time. The sample was heated under air in a Paar HTK temperature chamber. From 313 to 1273 K, diffractograms were taken every 10 K. Heating rates of 30 K/min were applied. It is important to note that, due to the time needed to accumulate the data, the effective heating rate over the whole temperature range is only 0.1 K/min. Analysis of the PXRD data was performed using the Diffraction Plus EVA (Version 9.0) and Topaz software (Version 4.2). Thermal expansion and quantitative phase analysis was done with the Rietveld method. Structural phase parameters, isotropic temperature factors, lattice parameters, size-like Lorentzian peak broadening, and scale

factors were refined independently for both phases. No corrections for systematic peak shifts were necessary. Starting values for the structures were taken from the literature.^{6,22} With this, two lattice parameters and eight parameters with influence on the phase fraction are calculated from 28 peaks for the hexagonal phase. For the tetragonal phase, two lattice and five fraction-influencing parameters are calculated from 10 peaks. This results in accuracies of ± 0.0005 Å and ± 0.5 wt %, respectively.

2.3. Raman Spectroscopy. High-temperature confocal Raman spectra were obtained with a Horiba Jobin Yvon LabRam-HR 800 Raman microspectrometer. The samples were excited by the 532 nm emission line of a 100 mW Nd:YAG laser. The size and power of the laser spot on the surface were approximately 1 μm and 5 mW, respectively. A grating with 1800 lines/mm was used. The spectral resolution, determined by measuring the Raleigh line, was about 1.4 cm^{-1} . The dispersed light was collected by a 1024×256 open electrode CCD detector. The confocal pinhole and spectrometer entrance slits were set to 1000 and 100 μm , respectively. Spectra were recorded unpolarized. Accuracy of Raman line shifts in the order of 0.5 cm^{-1} was achieved by adjusting the zero-order position of the grating and regularly checked by measuring a neon spectral calibration lamp. Baseline and band fitting of the spectra was done by the LabSpec 5 software, assuming a second-order polynomial background and folded Gaussian–Lorentzian peak shape. The high-temperature Raman measurements were performed using a water-cooled Linkam TS1500 heating stage from room temperature up to about 1260 K. The temperature regions between 373 and 573 K, as well as above 1123 K, were investigated in detail and Raman spectra taken every 20–30 K. In between, spectra were taken in 50–100 K steps. The samples were usually heated to the required temperature with a heating rate of 50 K/min and the spectra taken immediately after reaching the desired temperature. At 1260 K, also time-resolved Raman measurements were conducted.

2.4. Differential Thermal Analysis. Thermal analysis was conducted on a SETSYS EVOLUTION 2400 TG-DTA 1600 $^\circ\text{C}$, using the Calisto software. About 25 mg of sample was placed in an open corundum container of 4 mm diameter and heated from 298 K up to 1473 K at a rate of 5 K/min in a helium atmosphere. Simultaneous mass spectrometry analysis was conducted on a Pfeiffer Omnistar GSD 301 quadrupole mass spectrometer for all masses in the range 8–90 au, with a resolution of 1 au and 1 s measuring time for each mass. The temperature and energy calibration was performed using In, Sn, Pb, Zn, Al, Ag, Au, and N.

3. RESULTS AND DISCUSSION

Recent experiments by some of the authors showed the unexpected formation of a small amount of tetragonal GeO_2 following transformation from the corresponding hexagonal phase upon annealing treatments at 673 K in various gas atmospheres (He , O_2 , H_2), eventually spoiling the use of the latter as, e.g., oxide support for small noble metal catalyst particles.¹⁴ Thus, high-resolution XRD diffraction patterns were taken in 10 K steps between 300 and 1300 K for both the pure hexagonal and the RbOH-impregnated hexagonal phase in order to further investigate this effect. From the XRD diffractograms, information on the amount of the participating phases (tetragonal and hexagonal), the corresponding changes in the lattice parameters, and the change in the unit cell volume of the hexagonal phase

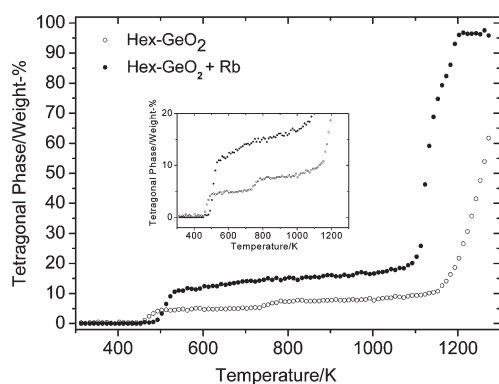


Figure 1. Amount of tetragonal phase (weight %) as a function of the annealing temperature between 313 and 1300 K for pure hexagonal and RbOH-impregnated hexagonal GeO_2 as deduced from the high-resolution and -temperature X-ray diffraction spectra.

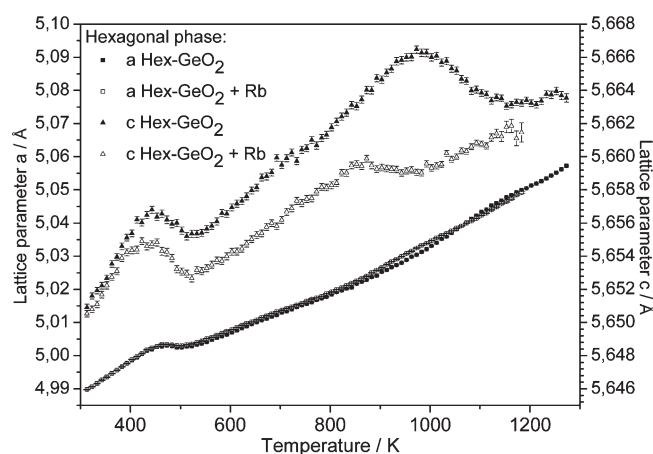


Figure 2. Evaluation of the changes of the lattice parameters a and c of the hexagonal GeO_2 phases with and without RbOH impregnation upon annealing between 313 and 1300 K.

were extracted, as outlined in detail in the Experimental Section and the Supporting Information (Figures S1a and S1b).

Figure 1, highlighting the amount of the tetragonal phase as a function of the annealing temperature for both the hexagonal and the RbOH-impregnated hexagonal phase, summarizes the peculiar phase transformation behavior between the two polymorphic GeO_2 structures. It is immediately obvious that the transformation into the tetragonal phase is not a one-step process but, for both samples, proceeds via clearly discernible steps. On pure hexagonal GeO_2 , transformation of a small part (up to about 5%) of the hexagonal into the tetragonal phase starts already at 450 K and is finished at around 510 K. Between 510 and 730 K, further transformation nearly stops and the amount of the tetragonal phase increases only very little with the annealing temperature. A second transformation step is observed at 730 K, resulting in a total amount of $\sim 8\%$ tetragonal phase. Above 730 K, the amount of tetragonal phase again stays nearly constant up to 1150 K, above which the formation of the tetragonal phase is strongly accelerated. It is worth noting that even at the highest annealing temperatures (~ 1300 K) no complete transformation into the tetragonal phase could be achieved.

A very similar overall reaction profile has been obtained on the RbOH-impregnated hexagonal GeO_2 sample, with a few notable

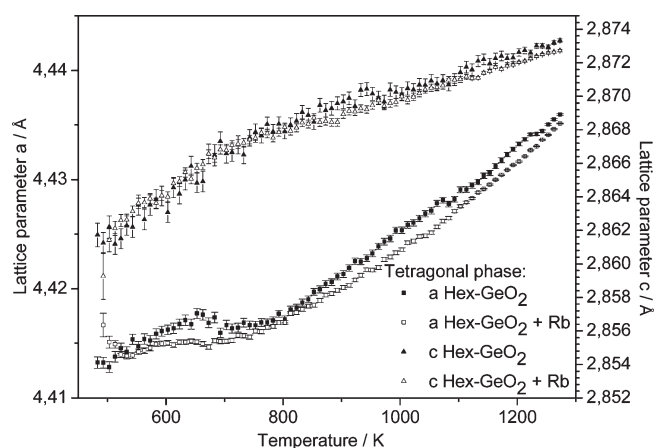


Figure 3. Evaluation of the changes of the lattice parameters a and c of the tetragonal GeO_2 phase formed by annealing the hexagonal GeO_2 structure with and without RbOH impregnation between 313 and 1300 K.

exceptions. It appears that the first phase transformation is slightly shifted to higher temperatures (~ 470 K) and that, in between ~ 500 and 1100 K, the amount of tetragonal phase is considerably higher at $\sim 12\%$. Moreover, in this temperature range, the amount of tetragonal phase smoothly increases, but the small step at ~ 730 K observed on the undoped sample is absent. Instead, a small reduction of the rate of phase increase takes place in the same temperature region. Again, strong acceleration of tetragonal phase formation is observed at high temperatures. In contrast to pure hexagonal GeO_2 , this last step already starts at ~ 1070 K (i.e., at ~ 100 K lower temperatures as compared to pure hexagonal GeO_2) and complete transformation into the tetragonal phase occurs already at 1170 K.

Although the high-temperature behavior of both pure hexagonal and the alkali-doped GeO_2 has been scrutinized,^{1,20,21,23,24} it is not understood what happens during the phase transformation starting at around 470 K, why this partial phase transformation takes place at all, and why it stops all of a sudden. It is therefore tempting to analyze the changes in the crystal lattice structure in this temperature range. Figures 2 and 3 correspondingly show the changes in the lattice parameters a and c of the hexagonal and tetragonal phases, both for the pure and the RbOH-impregnated sample. Focusing on the changes in the lattice parameters of the hexagonal phase (Figure 2), some peculiarities are notable. Up to 470 K, both lattice parameters (a and c) increase (i.e., the lattice expands), as anticipated from a conventional thermal expansion behavior. In contrast but coinciding well with the onset of the first phase transformation step at ~ 450 K discussed in Figure 1, the lattice contracts until this phase transformation is finished (up to ~ 510 – 520 K), after which a normal annealing behavior is observed up to 970 K. The second phase transformation step can only be seen as a small bump at ~ 730 K. Interestingly, another substantial lattice contraction is observed between 730 and 1130 K, that is, just before the final strongly accelerated formation of the tetragonal phase. The first lattice contraction can also be well observed in the lattice parameter a , which, however, seems not to be substantially affected by the final phase transformation at ~ 1130 K. This behavior is in agreement with neutron diffraction results of Haines et al., who interpreted the anisotropic lattice distortion upon annealing in terms of an increase in the inter-tetrahedral Ge–O–Ge bridging angle and a corresponding decrease in the tilt

angle of the GeO_4 tetrahedrons with respect to a hypothetical β -quartz-type structure.⁷ On the basis of the high degree of structural distortion up to the melting point, Haines et al. ruled out the possible occurrence of a phase transition for GeO_2 similar to the α – β quartz transformation.⁷ Hence, the phase transformation discussed in Figure 1 solely reflects a transition between the hexagonal and the tetragonal phase. Qualitatively similar lattice distortion behavior of the hexagonal structure can also be identified for the RbOH -impregnated sample. It nevertheless appears that the RbOH -impregnated hexagonal GeO_2 phase is generally less affected by the lattice distortions along its c axis and that the second lattice contraction is shifted to about 100 K lower temperature (~ 870 K). It should be emphasized at this point that also this second lattice contraction at higher temperatures can be linked to the onset of the final phase transformation to the tetragonal phase. The change in the lattice parameter is almost unaffected by the presence of RbOH . Since it has been reported that the introduction of a small amount of alkali oxides gives rise to the presence of GeO_6 octahedra and nonbridging oxygen species, subsequently leading to densification of the former hexagonal GeO_2 structure,¹³ the lattice contraction is intimately connected with the partial phase transformation into the tetragonal structure (the density of the tetragonal phase is 6.239 g cm^{-3} , the one of the hexagonal modification 4.228 g cm^{-3}).²⁴ This will again be discussed in connection with the Raman measurements (see below).

Concerning the alterations of the lattice parameters of the tetragonal phase (Figure 3), the associated changes appear to be more complex, since the tetragonal phase is only formed at and above 450 K. A less pronounced lattice contraction is observed for its lattice parameter a at around 673 K, both with and without RbOH . Above 770 K, anticipated temperature expansion behavior is observed. Normal temperature dependence has also been observed for the corresponding c parameter. Finally, it should be mentioned that the associated lattice expansions and contractions for both phases are also visible in the respective unit cell volumes (see Supporting Information, Figure S2).

Focusing especially on the phase transformations at around 500 K, it is already clear that they are accompanied by considerable anisotropic structural rearrangements. This of course immediately implies the question of the driving force for this phase transformation. The presence of RbOH is of some influence, but the phase transformation also occurs to a lower extent without RbOH . To answer this question, we applied spectroscopic (Raman) and thermal analysis (thermal gravimetric analysis coupled with mass spectrometry) techniques to the pure and RbOH -impregnated samples in order to eventually gain insight into the chemical nature of this phase transition.

Figure 4 highlights thermal gravimetric runs from 313 to 1473 K again for both GeO_2 samples. The graphs show both mass and heat change upon annealing (panel A) and, therefore, phase transformation. Both samples suffer a considerable weight loss (about 1.5–1.6 mass %) between ~ 470 and 630 K, i.e., in exactly the same temperature region, where the first phase transformation has been observed. Due to the fact that the phase transformations proceed over a broader temperature region, only very broad heat flow signals without pronounced features are expected, which basically is corroborated in Figure 4. Only at temperatures beyond 1273 K, partial retransformation of the tetragonal into the hexagonal phase and, at even higher temperatures (~ 1400 K), melting of the tetragonal GeO_2 modification give rise to rather sharp and well-defined endothermic heat signals.

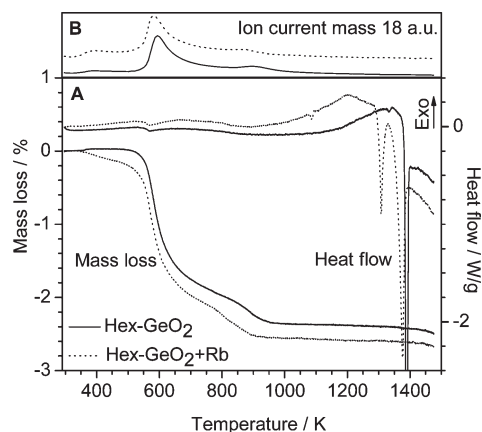


Figure 4. Thermal gravimetric analysis and heat flow upon annealing pure and RbOH -impregnated hexagonal GeO_2 between 313 and 1473 K (panel A). Panel B shows the corresponding mass spectrometer signals of mass $m/z = 18$.

The in-parallel monitored mass spectrometry signals, highlighted in panel B of Figure 4, finally reveal that the weight loss is due to removal of water trapped inside the crystal structure of hexagonal GeO_2 . In the case of pure hexagonal GeO_2 , two water peaks are visible, coinciding in temperature with the two phase transformations at ~ 470 and ~ 730 K, as observed in the XRD spectra (see upper panel of Figure 4). For RbOH -impregnated hexagonal GeO_2 , the first water peak appears unchanged, while the small second peak is less pronounced and tends to blend into the continuous water loss.

Taking the results from XRD, thermal gravimetric analysis, and mass spectrometer signals together, it is clear that (i) a limited part of the hexagonal GeO_2 with or without alkali impregnation is transformed at temperatures as low as ~ 470 K and (ii) this phase transformation is accompanied by both substantial lattice distortions and the loss of a considerable amount of water. From the fact that water is only removed at temperatures between 470 and 630 K, we deduce that the water is in fact more strongly bound either inside or as strongly bound surface OH groups.

To consolidate the crucial influence of water on the phase transformation at low temperatures and to eventually gain insight into the bonding of water to the Ge–O entities in the crystal lattice, temperature-dependent Raman measurements have been additionally performed as a spectroscopic counterpart to the XRD experiments discussed above. Regarding this phase transformation, Raman spectroscopy has only been intensively exploited to study the formation of rubidium germanate glasses at high temperatures¹³ and to elucidate thermodynamic properties at high temperatures.²⁴ It is especially powerful, since the Raman spectra of both GeO_2 modifications differ significantly and allow careful spectroscopic monitoring of the phase transition.¹ Both ex situ and in situ high-temperature Raman spectra have been recorded.

Figure 5A, among others, shows a typical Raman spectrum of the initial hexagonal GeO_2 sample taken at 298 K, which very well coincides with literature-reported spectra.^{1,4,23,24} It is, however, worth noting that, although all spectra reported in the literature exhibit the same features as dictated by the symmetries of the hexagonal crystal lattice, they differ by one notable exception: whereas a small peak at $\sim 780 \text{ cm}^{-1}$ (sometimes also a peak with

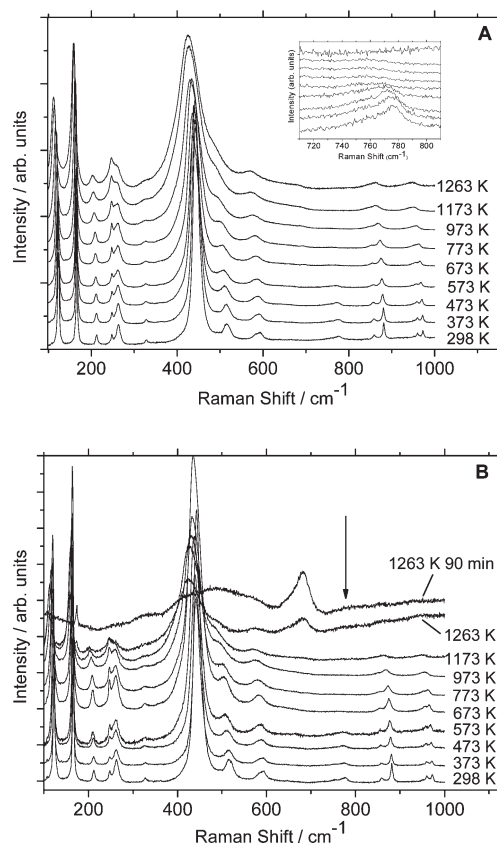


Figure 5. Selected temperature-dependent Raman spectra collected between 313 and 1300 K on pure hexagonal GeO_2 (A) and the RbOH-impregnated hexagonal GeO_2 sample (B). In panel A, the thermal evolution of the Raman bands between 740 and 780 cm^{-1} is shown as an inset.

a low-frequency shoulder at $\sim 760 \text{ cm}^{-1}$) is present in the studies of, e.g., Atuchin et al.⁴ or Gillet et al.,²⁴ this Raman feature is absent in the spectra provided, e.g., by Mernagh et al.²³ or Scott et al.¹ It should be emphasized that this small Raman peak does not arise from the hexagonal GeO_2 lattice, as shown by ab initio calculated Raman frequencies (see Supporting Information, Figure S3). Moreover, this peak gets progressively smaller upon annealing and completely vanishes upon heating to about 673 K, i.e., in the same temperature region where the partial phase transformation to tetragonal GeO_2 takes place and water is removed from the hexagonal sample (see the inset of the corresponding frequency region in Figure 5A). It is therefore tempting to assign this Raman feature to water bound to Ge–O entities. Following arguments that the Raman spectrum of GeO_2 can be conceptually divided into frequency regions where Raman features arise due to rotations and translations of GeO_4 tetrahedra (below 300 cm^{-1}), Ge–O bending vibrations (300–350 cm^{-1}), Ge–Ge stretching vibrations (up to 590 cm^{-1}), and finally Ge–O stretching vibrations within the GeO_4 tetrahedra (above 590 cm^{-1}), we assume that this peak might arise from the Ge–O stretching vibration of a water- or hydroxyl-distorted GeO_4 entity. This argument can also be chemically taken further if one considers the propensity of the hexagonal GeO_2 to partially dissolve in water ($\sim 0.4 \text{ g}/100 \text{ g}$ water at 298 K²⁵) and the possible resulting hydrated germanic acid related structures.^{17,25} It has been reported that GeO_2 forms monogermanic acid

(H_2GeO_3) in water.²⁴ Although unclear up to now, the solution possibly also includes $\text{Ge}(\text{OH})_4$ structures (i.e., $\text{H}_2\text{GeO}_3 \cdot \text{H}_2\text{O}$) and/or higher polyacidic adducts such as pentagermanic acid ($\text{H}_2\text{Ge}_5\text{O}_{11}$). The latter exhibits a bipyramidal structure, whose edges are occupied by five Ge and bridged by nine O atoms. The OH groups are connected to the terminal Ge centers.²⁵ In summary, in solution, GeO_2 forms germanates (i.e., negatively charged Ge–O entities), which are structurally similar to silicate compounds.^{17,25} Although salts of these germanic acids are well-known, the corresponding acids are not defined compounds, since in solution they are only observed as mixtures of acids of different condensation degrees. The same germanate compounds can also be prepared by melting GeO_2 and oxides (e.g., alkali-oxides or oxides of the second main group of the periodic table) together.²⁵ Most important, these germanate compounds exhibit Raman spectra that are distinctly different from those of the pure GeO_2 materials. Accordingly, the Raman features between ~ 740 and 780 cm^{-1} can be interpreted by correlating them to similar features in other germanate compounds, more specifically to a GeO_2 – Rb_2O germanate glass. For higher Rb_2O loadings ($x \geq 0.30$), the Raman spectra show additional features at around 740–765 cm^{-1} which are attributed to the symmetric vibration of Ge– O^- bonds (O^- denoting a nonbridging oxygen atom) in germanate tetrahedra.¹³ In analogy, we assign these Raman features also to the Ge–O stretching vibration of a germanate entity formed by heterolytically adsorbed water at a GeO_4 tetrahedral building block. Qualitatively very similar spectra have been obtained for the RbOH-impregnated GeO_2 sample (shown in Figure 5B; the corresponding water-related peaks are denoted by an arrow).

Above an annealing temperature of 673 K, the Raman spectra of the pure hexagonal sample are consistent with literature-reported spectra and result in the already well-known high-temperature effects; that is, peak shifts and peak broadening are observed (see also Figure 5A). More interesting, however, are similar Raman experiments on the RbOH-impregnated hexagonal GeO_2 sample (Figure 5B), which yielded quite similar results up to 1173 K. However, increasing the temperature to 1203 K resulted in the sudden appearance of the A_{1g} peak of the tetragonal GeO_2 structure.¹ Increasing the annealing temperature to 1263 K yields a higher amount of tetragonal GeO_2 in accordance with the XRD results, although some very broad peaks of the hexagonal modification are still present. Isothermal treatments at 1263 K for 90 min remove the rest of the hexagonal phase, and a Raman spectrum of the tetragonal structure with thermally broadened peaks and a large background is obtained. The corresponding spectrum after cooling to room temperature coincides well with a typical Raman spectrum of the tetragonal GeO_2 phase. The disappearance of the broad peak of hexagonal GeO_2 after 90 min clearly demonstrates the slow succession of the reaction. Taking into account the much higher heating rates of 50 K/min of the Raman experiments in contrast to only 0.1 K/min effective rate for the XRD measurements, both methods reveal consistently complete transformation of hexagonal to tetragonal GeO_2 at 1263 K.

In summary, the Raman experiments therefore support the XRD measurements. This scenario is also backed up by ex situ Raman spectra taken after different annealing treatments, which exclusively show tetragonal GeO_2 whenever a RbOH-impregnated hexagonal GeO_2 sample is annealed to 1263 K.

To put our results into literature perspective, it should be noted that this phase transformation is on the research focus for

several decades now. It is well-known that the transition from the hexagonal to the tetragonal phase can be triggered by high pressures and temperatures (2–40 kbar, 653–1100 K) and that the rate of transformation can be significantly accelerated by adding alkali compounds.^{11,20,21} This is usually done by mixing the hexagonal powder with alkali carbonates as “mineralization catalysts” (Li_2CO_3 , Na_2CO_3 , K_2CO_3 , or Rb_2CO_3) and subsequent annealing at high temperatures ($T > 1000$ K) to induce the formation of alkali germanate glasses.^{11,13,20,21} These germanate glasses are proven to represent the least activated intermediate of the structural rearrangement from the tetrahedral oxygen-coordinated Ge in hexagonal GeO_2 to the octahedral oxygen coordination in tetragonal GeO_2 .^{11,20,21} This change in coordination sphere might also proceed without breaking the Ge–O–Ge bond and, depending on the alkali dopant used, either in the solid (for Li or Na) or liquid (for K) state.^{20,21} This aspect of the phase transformation has been given a fair amount of attention in the literature.^{11,13,20,21} However, it has also been stated that this transformation does not occur without dopant in the similar temperature region, which, as deducible from the presented results, is only true to the point that no *complete* transformation can be achieved. The same conclusions can in principle be drawn from the experiments of Bertini et al., who observed a complete transformation to the tetragonal GeO_2 phase by annealing the hexagonal phase for 700 h at around 1023 K.¹⁵ In turn, all these facts infer a kinetically controlled nucleation scenario at least at high annealing temperatures. The situation, however, appears to be more complex, since (i) at least a small part of the hexagonal GeO_2 can be transformed to the tetragonal phase even at temperatures as low as 450 K, obviously under involvement of water, and (ii) this transformation seems to not be substantially dependent on the use of an alkali dopant. The influence of the dopant therefore becomes only apparent at high annealing temperatures, where the formation of the alkali germanate glass accelerates the subsequent change in the oxygen coordination sphere. Focusing on the transformation at lower temperatures, it has been repeatedly stated that pure hexagonal GeO_2 can be partly converted to the tetragonal form by, e.g., slow evaporation of a diluted GeO_2 solution (a maximum of 15% tetragonal GeO_2 has been reported to be obtainable)²⁵ or applying high pressures (10–30 kbar) at 653 K. Concerning the latter, an explanation of the inhibition of further growth of the tetragonal phase at 10–30 kbar has been given in terms of the initial formation of the rutile nucleation sites of the tetragonal phase at grain surfaces/interfaces of hexagonal GeO_2 and the subsequent formation of internal strain in the hexagonal GeO_2 grains hindering further phase transformation.¹⁷ Recent experiments by Balitsky et al. also indicate the existence of an α -quartz to rutile-like structure transformation at temperatures as low as 465 K for α -quartz GeO_2 single crystals grown in aqueous medium.²⁶ Interestingly, complete transformation into the tetragonal phase at 653 K has also been reported by the use of crystallization aids such as ammonium fluoride but only in case the diluted hexagonal GeO_2 solution has been annealed to 653 K prior to impregnation with ammonium fluoride. Repeated cycles of annealing to 653 K and impregnation eventually caused complete transformation. A strong dependence of the pretreatment has therefore been observed. Below 653 K, under no experimental conditions did a phase transformation take place.¹⁹ We, however, unambiguously show that this transformation already starts at 450 K (with a further small transformation at ~ 730 K in the case of pure hexagonal GeO_2), but surprisingly, no further transformation

takes place up to at least 1150 K. This behavior is interpreted in terms of the formation of germanate entities upon reaction of hexagonal GeO_2 with water (possibly trapped during the manufacturing process), which decomposes upon annealing to ~ 470 K and (at least partially) structurally reorganizes the Ge–O lattice to the tetragonal GeO_2 phase. When all water is released, further transformation stops until at high temperatures the reaction obviously very much depends on the kinetics of the nucleation and growth of tetragonal GeO_2 . In the alkali-impregnated case, the reaction further proceeds and is catalyzed by alkali germanate glasses. It should be strongly emphasized that both phase transformations proceed over similar germanate compounds, albeit including water or rubidium oxide and, therefore, at considerably different temperatures.

4. CONCLUSIONS

In conclusion, it has been shown that the phase transformation between the hexagonal and the tetragonal GeO_2 modification is much more complex than previously assumed, especially in the case of pure hexagonal GeO_2 . The removal of water from a hexagonal sample partially converted to germanate entities has been identified as the crucial reaction step in distorting the crystal lattice and triggering the phase transformation at low temperatures (~ 430 K). Although the exact mechanism is unclear yet, the results imply that water is able to alter the tetrahedral coordination sphere of Ge in the hexagonal GeO_2 lattice to germanate-related building blocks in a similar way as alkali dopants do at high temperatures. Obviously, the annealing behavior of hexagonal GeO_2 crucially depends on the manufacturing prehistory and in case GeO_2 is prepared from aqueous solution (e.g., by hydrolysis of GeCl_4) will inevitably give rise to the formation of residual germanic acid species and consequently the peculiar partial phase transformation at ~ 470 K to the tetragonal modification is induced.

■ ASSOCIATED CONTENT

S Supporting Information. (A) High-resolution and-temperature plots of pure hexagonal and RbOH-impregnated hexagonal GeO_2 (T -range 40–1000 °C corresponding to 313–1273 K). (B) Temperature dependence of the unit cell volumes of the hexagonal and tetragonal phases. (C) Comparison of the measured and ab initio calculated Raman frequencies of the hexagonal GeO_2 phase. This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: simon.penner@uibk.ac.at. Phone: 00435125075056. Fax: 00435125072925.

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REFERENCES

- (1) Scott, J. F. *Phys. Rev. B* **1970**, *1* (8), 3488–3493.
- (2) Ardyanin, M.; Rinnert, H.; Vergnat, M. *J. Lumin.* **2009**, *129*, 729–733.
- (3) Terakado, N.; Tanaka, K. *J. Non-Cryst. Solids* **2006**, *352*, 3815.
- (4) Atuchin, V. V.; Gavrilova, T. A.; Gromilov, S. A.; Kostrovsky, V. G.; Pokrovsky, L. D.; Troitskaya, I. B.; Vemuri, R. S.; Carbajal-Franco, G.; Ramana, C. V. *Cryst. Growth Des.* **2009**, *9* (4), 1829. Ramana, C. V.; Carbajal-Franco, G.; Vemuri, R. S.; Troitskaya, I. B.; Gromilov, S. A.; Atuchin, V. V. *Mater. Sci. Eng., B* **2010**, *174*, 279–284 and references therein.
- (5) Bolzan, A. A.; Fong, C.; Kennedy, B. J.; Howard, C. J. *Acta Crystallogr., Sect. B* **1997**, *53*, 373–380.
- (6) Baur, W. H.; Khan, A. A. *Acta Crystallogr., Sect. B* **1971**, *27*, 2133–2139.
- (7) Haines, J.; Cambon, O.; Phillipot, E.; Chapon, L.; Hull, S. J. *Solid State Chem.* **2002**, *166*, 434–441.
- (8) Prakapenka, V. P.; Shen, G.; Dubrovinsky, L. S.; Rivers, M. L.; Sutton, S. R. *J. Phys. Chem. Solids* **2004**, *65*, 1537–1545.
- (9) Hollemann, A. F.; Wiberg, E. *Lehrbuch der Anorganischen Chemie*; Walter deGruyter: Berlin, 2007.
- (10) Sarver, J. F.; Hummel, F. A. *J. Electrochem. Soc.* **1961**, *108*, 195–196.
- (11) Murthy, M. K.; Ip, J. *Nature* **1964**, *201*, 285–286.
- (12) Kotera, Y.; Yonemura, M. *J. Am. Ceram. Soc.* **1969**, *52*, 210–215.
- (13) Kamitsos, E. I.; Yiannopoulos, Y. D.; Karakassides, M. A.; Chrysikos, G. D.; Jain, H. *J. Phys. Chem.* **1996**, *100*, 11755–11765.
- (14) Zhao, Q.; Lorenz, H.; Turner, S.; Lebedev, O. I.; VanTendeloo, G.; Rameshan, C.; Klötzer, B.; Konzett, J.; Penner, S. *Appl. Catal., A* **2010**, *375*, 188–195.
- (15) Bertini, L.; Ghigna, P.; Scavini, M.; Cargoni, F. *Phys. Chem. Chem. Phys.* **2003**, *5*, 1451–1456.
- (16) Kotera, Y.; Yonemura, M. *Trans. Faraday Soc.* **1963**, *59*, 147–152.
- (17) Zeto, R. J.; Roy, R. *Reactiv. Solids, 6th Proc. Int. Symp.* **1969**, 803–815.
- (18) Ault, K. M.; Secco, R. A. *Solid State Commun.* **1996**, *98*, 449–452.
- (19) Schwarz, R.; Haschke, E. *Z. Anorg. Chem.* **1943**, *252*, 170–172.
- (20) Yonemura, M.; Kotera, Y. *Bull. Chem. Soc. Jpn.* **1974**, *47* (4), 789–792.
- (21) Yonemura, M.; Kotera, Y. *Bull. Chem. Soc. Jpn.* **1974**, *47* (4), 793–796.
- (22) Jorgensen, J. D. *J. Appl. Phys.* **1978**, *49*, 5473.
- (23) Mernagh, T. P.; Liu, L. *Phys. Chem. Miner.* **1997**, *24*, 7–16.
- (24) Gillet, P.; Le Cleac’h, A.; Madon, M. *J. Geophys. Res.* **1990**, *95*, 21635–21655.
- (25) Meyer, R. J. *Gmelins Handbuch der Anorganischen Chemie*, Deutsche Chemische Gesellschaft, Band 45 including Ergänzungsband, Weinheim, 1958.
- (26) Balitsky, D. V.; Balitsky, V. S.; Pisarevsky, Y. V.; Philippot, E. O.; Vestrova, Y.; Pushcharovsky, D. Y. *Ann. Chim. Sci. Mater.* **2001**, *26*, 183.